# Review

Z. Song

## Section 1.4 Linear subspace

What is a linear subspace?

Operations between subspaces:

sum, intersection, direct sum, complementary subspace

## Section 1.5 Linear transformation

linear mapping from one space to another,

associated matrix representation,

the properties of linear transformation, ect…

- What is linear transformation? (LP)
- Matrix representation, representation for LP transformation

# Outline



- Section 1.5 Linear transformation
- Notion
- Matrix representation
- Isomorphism
- Range and kernel
- Composite transformation

- Chapter 2- Polynomials and matrix polynomials
  - Part 1. Polynomials –the first four sections
  - Part 2. Matrix polynomials –the last four sections
- Today we go into details of Section 2.1 -Section 2.2
  - S1. Linear algebras
  - S2. Polynomial ring and Euclidean division

# 1.5 Linear transformation

Z. Song

·Linear transformation ---function mapping elements in a linear space to another

Definition 1.21 Let $\mathcal{U}$ and $\mathcal{V}$ be linear spaces over a field $\mathcal{F}$, a mapping $\mathscr{A} : \mathcal{U} \mapsto \mathcal{V}$ is said to be a linear transformation from $\mathcal{U}$ to $\mathcal{V}$ if it is additive and homogeneous, that is, for any two elements $\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_2$ in $\mathcal{U}$ and any scalar $k$ in $\mathcal{F}$

$$\mathscr{A}(\boldsymbol{\alpha}_1 + \boldsymbol{\alpha}_2) = \mathscr{A}(\boldsymbol{\alpha}_1) + \mathscr{A}(\boldsymbol{\alpha}_2),$$
$$\mathscr{A}(k\boldsymbol{\alpha}_1) = k\mathscr{A}(\boldsymbol{\alpha}_1)$$

$\mathscr{A}(\boldsymbol{\alpha}) \in \mathcal{V}$ is called the image of $\boldsymbol{\alpha} \in \mathcal{U}$ under $\mathscr{A}$. The totality of all images of the elements in $\mathcal{U}$ is denoted

$$\mathscr{A}(\mathcal{U}) \triangleq \{\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathcal{V} \text{ and } \boldsymbol{\beta} = \mathscr{A}(\boldsymbol{\alpha}) \text{ for some } \boldsymbol{\alpha} \in \mathcal{U}\}$$

and is referred to as the image (or the range) of $\mathcal{U}$ under $\mathscr{A}$. $\mathcal{U}$ is the domain of $\mathscr{A}$.

# 1.5 Linear transformation



- Identity transformation and zero transformation

Exp: A transformation $\mathscr{E} : \mathcal{V} \mapsto \mathcal{V}$ is defined as

$$\mathscr{E}(\boldsymbol{\alpha}) = \boldsymbol{\alpha} \in \mathcal{V} \ \forall \ \boldsymbol{\alpha} \in \mathcal{V}$$

It can be readily checked that $\mathscr{E}$ is a linear transformation. $\mathscr{E}$ is called the identity transformation.

Exp: Let $\mathscr{O} : \mathcal{U} \mapsto \mathcal{V}$ be defined as

$$\mathscr{O}(\boldsymbol{\alpha}) = \mathbf{0} \ \forall \ \boldsymbol{\alpha} \in \mathcal{U}$$

Then it can be readily checked that $\mathscr{O}$ is a linear transformation. Since $\mathscr{O}$ transforms all elements in $\mathcal{U}$ identically into $\mathbf{0} \in \mathcal{V}$, it is called the zero transformation.

# 1.5 Linear transformation

Z. Song

- Properties of linear transformation

  A linear transformation has the following properties.

  1. $\mathscr{A}(\mathbf{0}) = \mathbf{0}$;

  2. $\mathscr{A}\left(\sum_{i=1}^{s} k_i \boldsymbol{\alpha}_i\right) = \sum_{i=1}^{s} k_i \mathscr{A}(\boldsymbol{\alpha}_i)$;

  3. If $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s \in \mathcal{V}$ are linearly dependent, then the images $\mathscr{A}(\boldsymbol{\alpha}_1), \mathscr{A}(\boldsymbol{\alpha}_2), \ldots, \mathscr{A}(\boldsymbol{\alpha}_s)$ are also linearly dependent.

# 1.5 Linear transformation



- An example of linear transformation

Exp: Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

be an $m \times n$ real matrix. It can be readily checked that $\mathscr{A} : \mathcal{R}^n \mapsto \mathcal{R}^m$ defined as

$$\mathscr{A}(\boldsymbol{\alpha}) = A\boldsymbol{\alpha} \in \mathcal{R}^m \ \forall \ \boldsymbol{\alpha} \in \mathcal{R}^n$$

is a linear transformation from $\mathcal{R}^n$ to $\mathcal{R}^m$.

# 1.5 Linear transformation



- Remark on linear transformation

If the vectors $\alpha_1, \alpha_2, \ldots, \alpha_s \in \mathcal{U}$ are linearly dependent, the the images $\mathscr{A}(\alpha_1), \mathscr{A}(\alpha_2), \ldots, \mathscr{A}(\alpha_s)$ must be linearly dependent. However, the statement cannot be made in the opposite direction. Even though the vectors $\alpha_1, \alpha_2, \ldots, \alpha_s \in \mathcal{U}$ are linearly independent, their images $\mathscr{A}(\alpha_1), \mathscr{A}(\alpha_2), \ldots, \mathscr{A}(\alpha_s)$ may be linearly dependent, -see e.g. the zero transformation. is a linear transformation from $\mathcal{R}^n$ to $\mathcal{R}^m$.

# 1.5 Linear transformation



• Matrix representation of a linear transformation

For notational simplicity, we shall denote by $\mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$ the set of all linear transformations from the linear space $\mathcal{U}$ to the linear space $\mathcal{V}$ over the field $\mathcal{F}$. Let $[\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]$ be a basis of $\mathcal{U}$, and $[\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m]$ be a basis of $\mathcal{V}$, $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$ be a linear transformation, then the image of a basis vector in $\mathcal{U}$ can be represented as

$$\mathscr{A}(\boldsymbol{\alpha}_j) = \sum_{i=1}^{m} a_{ij}\boldsymbol{\beta}_i = [\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m] \underbrace{\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}}_{\triangleq \boldsymbol{a}_j}$$

the totality of the images of all the basis vectors can be written as

$$\begin{aligned} \mathscr{A}([\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]) &\triangleq [\mathscr{A}(\boldsymbol{\alpha}_1) \quad \mathscr{A}(\boldsymbol{\alpha}_2) \quad \dots \quad \mathscr{A}(\boldsymbol{\alpha}_n)] \\ &= \begin{bmatrix} \sum_{i=1}^{m} a_{i1}\boldsymbol{\beta}_i & \sum_{i=1}^{m} a_{i2}\boldsymbol{\beta}_i & \dots & \sum_{i=1}^{m} a_{in}\boldsymbol{\beta}_i \end{bmatrix} \\ &= [\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m] \underbrace{[\boldsymbol{a}_1 \quad \boldsymbol{a}_2 \quad \dots \quad \boldsymbol{a}_n]}_{\triangleq A} \end{aligned}$$

# 1.5 Linear transformation



- Matrix representation of a linear transformation

$A$ is referred to as the matrix representation of the linear transformation $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$ with respect to the bases $[\alpha_1 \quad \alpha_2 \quad \dots \quad \alpha_n]$ and $[\beta_1 \quad \beta_2 \quad \dots \quad \beta_m]$.

A direct use of the matrix representation is to evaluate the representation of the image $\mathscr{A}(\alpha)$ in relation to the representation of any $\alpha \in \mathcal{U}$.

- Matrix representation gives the relation of an element in a linear space and its images in another space.

# 1.5 Linear transformation

Z. Song

Let $\boldsymbol{\alpha} \in \mathcal{U}$, then

$$\boldsymbol{\alpha} = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n] \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{=\boldsymbol{x}}$$

- Coordinates transformation

The image $\mathscr{A}(\boldsymbol{\alpha}) \in \mathcal{V}$ can be represented as

$$\mathscr{A}(\boldsymbol{\alpha}) = [\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m] \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{=\boldsymbol{y}}$$

From

$$\begin{aligned} \mathscr{A}(\boldsymbol{\alpha}) &= \mathscr{A}([\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]\boldsymbol{x}) \\ &= \mathscr{A}(x_1\boldsymbol{\alpha}_1 + x_2\boldsymbol{\alpha}_2 + \cdots + x_n\boldsymbol{\alpha}_n) \\ &= \mathscr{A}(\boldsymbol{\alpha}_1)x_1 + \mathscr{A}(\boldsymbol{\alpha}_2)x_2 + \cdots + \mathscr{A}(\boldsymbol{\alpha}_n)x_n \\ &= [\mathscr{A}(\boldsymbol{\alpha}_1) \quad \mathscr{A}(\boldsymbol{\alpha}_2) \quad \dots \quad \mathscr{A}(\boldsymbol{\alpha}_n)]\,\boldsymbol{x} \\ &= [\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m]\,A\boldsymbol{x} \end{aligned}$$

and the uniqueness of the representation with respect to a given fixed basis, it follows that

$$\boldsymbol{y} = A\boldsymbol{x}$$

# 1.5 Linear transformation

Z. Song

Let $\boldsymbol{\alpha} \in \mathcal{U}$, then

$$\boldsymbol{\alpha} = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n] \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{=\boldsymbol{x}}$$

The image $\mathscr{A}(\boldsymbol{\alpha}) \in \mathcal{V}$ can be represented as

$$\mathscr{A}(\boldsymbol{\alpha}) = [\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m] \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{=\boldsymbol{y}}$$

From

$$\begin{aligned} \mathscr{A}(\boldsymbol{\alpha}) &= \mathscr{A}([\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]\boldsymbol{x}) \\ &= \mathscr{A}(x_1\boldsymbol{\alpha}_1 + x_2\boldsymbol{\alpha}_2 + \cdots + x_n\boldsymbol{\alpha}_n) \\ &= \mathscr{A}(\boldsymbol{\alpha}_1)x_1 + \mathscr{A}(\boldsymbol{\alpha}_2)x_2 + \cdots + \mathscr{A}(\boldsymbol{\alpha}_n)x_n \\ &= [\mathscr{A}(\boldsymbol{\alpha}_1) \quad \mathscr{A}(\boldsymbol{\alpha}_2) \quad \dots \quad \mathscr{A}(\boldsymbol{\alpha}_n)]\,\boldsymbol{x} \\ &= [\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m]\,A\boldsymbol{x} \end{aligned}$$

and the uniqueness of the representation with respect to a given fixed basis, it follows that

$$\boldsymbol{y} = A\boldsymbol{x}$$

# 1.5 Linear transformation



- The matrix representation of a linear transformation is unique when the bases are fixed.
- A matrix and a linear transformation is one to one correspondence when the bases are fixed.

> Theorem 1.12 Let $[\boldsymbol{\alpha}_1 \ \boldsymbol{\alpha}_2 \ \ldots \ \boldsymbol{\alpha}_n]$ be a basis of $\mathcal{U}$, and $[\boldsymbol{\beta}_1 \ \boldsymbol{\beta}_2 \ \ldots \ \boldsymbol{\beta}_m]$ be a basis of $\mathcal{V}$. Then for any given $m \times n$ matrix $A$, there exists a unique linear transformation $\mathscr{A}$ whose matrix representation with respect to the bases is $A$.

- The proof needs the concept of the sum of linear transformation.

# 1.5 Linear transformation



- Sum of linear transformation

Defintion 1.23 For $\mathscr{A}$, $\mathscr{B} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$, the sum transformation $(\mathscr{A} + \mathscr{B})$ of $\mathscr{A}$ and $\mathscr{B}$ is defined as

$$(\mathscr{A} + \mathscr{B})(\boldsymbol{\alpha}) = \mathscr{A}(\boldsymbol{\alpha}) + \mathscr{B}(\boldsymbol{\alpha}) \quad \text{for all } \boldsymbol{\alpha} \in \mathcal{U}.$$

Similarly, for $(k, \mathscr{A}) \in \mathcal{F} \times \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$, the scalar multiplication transformation $(k\mathscr{A})$ is defined as

$$(k\mathscr{A})(\boldsymbol{\alpha}) = k\mathscr{A}(\boldsymbol{\alpha}) \quad \text{for all } \boldsymbol{\alpha} \in \mathcal{U}.$$

With respect to the addition and scalar multiplication defined above, $\mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$ is a linear space with the zero element $\mathscr{O}$. The negative element of $\mathscr{A}$ is $-\mathscr{A}$, which can be seen from the following equations

$$\begin{aligned} (\mathscr{A} + (-\mathscr{A}))(\boldsymbol{\alpha}) &= \mathscr{A}(\boldsymbol{\alpha}) + (-\mathscr{A})(\boldsymbol{\alpha}) \\ &= \mathscr{A}(\boldsymbol{\alpha}) - \mathscr{A}(\boldsymbol{\alpha}) = \mathbf{0} \quad \forall \boldsymbol{\alpha} \in \mathcal{U} \end{aligned}$$

# 1.5 Linear transformation



- Sum of linear transformation

It is clear that the matrix representation of a sum transformation is the sum of the corresponding matrix representations, and the matrix representation of a scalar multiplication transformation is the scalar multiplication of the corresponding scalar and the matrix representation. The definition of sum transformation can be readily extended to a finite number of linear transformations $\mathscr{A}$, $\mathscr{B}$, $\mathscr{C}$, ..., $\mathscr{T} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$.

- Base on the concept of the sum of linear transformation, let us have a look at the proof the Theorem 1.12 on Page 24-25.

- Please have a look at Example 1.26 on page 25.

# 1.5 Linear transformation



•Given a pair of bases, the matrix representation of the linear transformation is fixed and unique, for another pair of bases, there is another matrix representation, what is the relationship of different matrix representations?

Theorem 1.13 Let $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$, $[\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]$ and $\left[\boldsymbol{\alpha}'_1 \quad \boldsymbol{\alpha}'_2 \quad \dots \quad \boldsymbol{\alpha}'_n\right]$ be two different bases of $\mathcal{U}$ with the transition matrix $P$, $[\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m]$ and $\left[\boldsymbol{\beta}'_1 \quad \boldsymbol{\beta}'_2 \quad \dots \quad \boldsymbol{\beta}'_m\right]$ be two different bases of $\mathcal{V}$ with the transition matrix $Q$. The matrix representation of $\mathscr{A}$ with respect to the bases $[\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]$ and $[\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m]$ is $A$, and the one with respect to the bases $\left[\boldsymbol{\alpha}'_1 \quad \boldsymbol{\alpha}'_2 \quad \dots \quad \boldsymbol{\alpha}'_n\right]$ and $\left[\boldsymbol{\beta}'_1 \quad \boldsymbol{\beta}'_2 \quad \dots \quad \boldsymbol{\beta}'_m\right]$ is $A'$. Then

$$A' = Q^{-1}AP\,.$$

# 1.5 Linear transformation



- Implications of Theorem 1.13, Equivalent transformation/matrix

For $n$-dimensional space $\mathcal{U}$ and $m$-dimensional space $\mathcal{V}$, with respect to a series different bases, $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$ has a series of different matrix representations $A$, $B$, ...

- Definition of equivalent matrix

Definition 1.24 Let $A$, $B \in \mathcal{F}^{m\times n}$. If there exists nonsingular matrices $Q \in \mathcal{F}^{m\times m}$ and $P \in \mathcal{F}^{n\times n}$ such that

$$B = QAP$$

then $B$ is said to be equivalent to $A$, written $B \cong A$.

# 1.5 Linear transformation



- Properties of equivalent matrices

It can be readily verified that

1. $A = E_m A E_n$, thus $A \cong A$ and "$\cong$" is reflexive,
2. $B = QAP, \iff A = Q^{-1}BP^{-1}$, thus $A \cong B$ and "$\cong$" is symmetric,
3. $B = QAP$, $C = RBS = (RQ)A(PS)$, thus $C \cong B$ and $B \cong A$ yield $A \cong A$, and "$\cong$" is transitive.

A nonsingular matrix can be decomposed into the product of some elementary matrices. Premultiplying $A$ by an elementary matrix corresponds to an elementary row manipulation, while post-multiplying $A$ by an elementary matrix corresponds to an elementary column manipulation. $B$ is equivalent to $A$, then $B$ is obtained from $A$ through a series elementary row and/or column manipulations.

# 1.5 Linear transformation



- Equivalent transformation/matrix

For $n$-dimensional space $\mathcal{U}$ and $m$-dimensional space $\mathcal{V}$, with respect to a series different bases, $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$ has a series of different matrix representations $A$, $B$, ... From this definition and Theorem 1.13, $A$, $B$, ... are equivalent matrices. On the other hand, equivalent $m \times n$ matrices represents also a single linear transformation with respect to different bases. The representation $[x_1 \quad x_2 \quad \dots \quad x_n]^{\mathsf{T}}$ of $\boldsymbol{\alpha}$ and the one $[y_1 \quad y_2 \quad \dots \quad y_m]^{\mathsf{T}}$ of its image $\mathscr{A}(\boldsymbol{\alpha})$ satisfy

$$y = Ax,$$

where $A$ is the matrix representaion of corresponding linear transformaion from $\mathcal{U}$ to $\mathcal{V}$.

# 1.5 Linear transformation



- Isomorphism on finite dimensional linear space

Definition 1.25 Let $\mathcal{U}$ and $\mathcal{V}$ be $n$- and $m$-dimensional linear spaces over $\mathcal{F}$, $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$. If for all $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2 \in \mathcal{U}$, $\boldsymbol{\alpha}_1 \neq \boldsymbol{\alpha}_2$ implies $\mathscr{A}(\boldsymbol{\alpha}_1) \neq \mathscr{A}(\boldsymbol{\alpha}_2)$, then $\mathscr{A}$ is injective. If for every $\boldsymbol{\beta} \in \mathcal{V}$ there exists $\boldsymbol{\alpha} \in \mathcal{V}$ such that $\boldsymbol{\beta} = \mathscr{A}(\boldsymbol{\alpha})$, $\mathscr{A}$ is called surjective. If $\mathscr{A}$ is both injective and surjective, then $\mathscr{A}$ is called a one-to-one correspondence (bijective).

Definition 1.26 Let $\mathcal{U}$, $\mathcal{V}$ be linear spaces over field $\mathcal{F}$. If the linear transformation $\mathscr{I}$ is a one-to-one correspondence between $\mathcal{U}$ and $\mathcal{V}$, then $\mathscr{I} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$ is referred to as an isomorphism. If there exists an isomorphism between $\mathcal{U}$ and $\mathcal{V}$ which maps $\mathcal{U}$ to $\mathcal{V}$, then we say that $\mathcal{U}$ is isomorphic with $\mathcal{V}$, denoted $\mathcal{U} \simeq \mathcal{V}$.

# 1.5 Linear transformation



- Properties of isomorphism

Obviously, the relation "$\mathcal{U} \simeq \mathcal{V}$" is an equivalence relation in the sense that

1. $\mathcal{V} \simeq \mathcal{V}$ (a linear space is isomorphic with itself, in this case $\mathscr{I}$ can be the identity transformation $\mathscr{E}$). This property is referred to as reflexivity;

2. If $\mathcal{U} \simeq \mathcal{V}$, then $\mathcal{V} \simeq \mathcal{U}$. This is the property of symmetry;

3. If $\mathcal{U} \simeq \mathcal{V}$, $\mathcal{V} \simeq \mathcal{W}$, then $\mathcal{U} \simeq \mathcal{W}$, that is, "$\simeq$" is transitive.

Because of the equivalence relation, we say also that $\mathcal{U}$ and $\mathcal{V}$ are isomorphic spaces.

# 1.5 Linear transformation

Z. Song

- Properties of isomorphism

Let $\mathcal{U}$ and $\mathcal{V}$ be linear spaces over field $\mathcal{F}$, $\mathscr{I} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$ is isomorphic, then

1. $\mathscr{I}(\boldsymbol{\alpha}) = \mathbf{0}$ if and only if $\boldsymbol{\alpha} = \mathbf{0}$;

2. The vectors $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_n$ in $\mathcal{U}$ are linearly independent, if and only if so are the images $\mathscr{I}(\boldsymbol{\alpha}_1), \mathscr{I}(\boldsymbol{\alpha}_2), \ldots, \mathscr{I}(\boldsymbol{\alpha}_n)$ in $\mathcal{V}$;

3. $\mathscr{I}$ maps a basis of $\mathcal{U}$ into a basis of $\mathcal{V}$, hence $\dim \mathcal{U} = \dim \mathcal{V}$;

4. If $\mathcal{U}_1$ is a subspace of $\mathcal{U}$, then

   $$\mathcal{V}_1 = \mathscr{I}(\mathcal{U}_1) = \{\, \boldsymbol{\beta} \;:\; \boldsymbol{\beta} = \mathscr{I}(\boldsymbol{\alpha}),\ \boldsymbol{\alpha} \in \mathcal{U}_1 \,\} \subset \mathcal{V}$$

   is a subspace of $\mathcal{V}$, and $\dim \mathcal{U}_1 = \dim \mathcal{V}_1$.

# 1.5 Linear transformation



- Necessary and sufficient conditions for isomorphism

Theorem 1.15 The two linear spaces $\mathcal{U}$ and $\mathcal{V}$ over a field $\mathcal{F}$ are isomorphic, if and only if $\dim \mathcal{U} = \dim \mathcal{V}$.

- Isomorphic spaces--- one to one correspondence, dimension remains the same, 'nothing is lost'.

# 1.5 Linear transformation



- Range and kernel of a linear transformation

Definition 1.27 Let $\mathcal{U}$ and $\mathcal{V}$ be linear spaces over $\mathcal{F}$. Let $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$. If we define

$$\mathscr{A}(\mathcal{U}) = \{\boldsymbol{\beta} \in \mathcal{V} \ : \ \exists\, \boldsymbol{\alpha} \in \mathcal{U} \text{ s.t. } \mathscr{A}(\boldsymbol{\alpha}) = \boldsymbol{\beta}\}$$

then the analysis reveals that $\mathscr{A}(\mathcal{U})$ is a subspace of $\mathcal{V}$.

If we define

$$\mathcal{N}(\mathscr{A}) = \{\boldsymbol{\alpha} \in \mathcal{U} \ : \ \mathscr{A}(\boldsymbol{\alpha}) = \mathbf{0}\}$$

then the analysis reveals that $\mathcal{N}(\mathscr{A})$ is a subspace of $\mathcal{U}$.

$\mathscr{A}(\mathcal{U})$ is called the range of $\mathscr{A}$ and denoted $\mathcal{R}(\mathscr{A})$. $\dim \mathcal{R}(\mathscr{A})$ is called the rank of $\mathscr{A}$, denoted $\text{rank}(\mathscr{A})$. $\mathcal{N}(\mathscr{A})$ is called the kernel of $\mathscr{A}$, denoted $\ker\mathscr{A}$, $\dim \mathcal{N}(\mathscr{A})$ is called the nullity of $\mathscr{A}$.

# 1.5 Linear transformation



- A theorem relates to the range and kernel of a linear transformation

Theroem 1.16 Let $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$, $\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_2$, ..., $\boldsymbol{\alpha}_n$ be a basis of $\mathcal{U}$, $\boldsymbol{\beta}_1$, $\boldsymbol{\beta}_2$, ..., $\boldsymbol{\beta}_m$ be a basis of $\mathcal{V}$, $A$ be the matrix representation of $\mathscr{A}$ with respect to these bases. Then

1. $\mathcal{R}(\mathscr{A}) = \text{span}\{\mathscr{A}(\boldsymbol{\alpha}_1), \mathscr{A}(\boldsymbol{\alpha}_2), \ldots, \mathscr{A}(\boldsymbol{\alpha}_n)\}$;

2. $\dim \mathcal{N}(\mathscr{A}) + \dim \mathcal{R}(\mathscr{A}) = n(= \dim \mathcal{U})$;

3. $\text{rank}(\mathscr{A}) = \text{rank}(A)$.

# 1.5 Linear transformation



- Composite transformation

  - Definition

Let $\mathcal{U}$, $\mathcal{W}$ and $\mathcal{V}$ be linear spaces over the field $\mathcal{F}$, $\mathscr{A} \in \mathcal{L}(\mathcal{W} \mapsto \mathcal{V})$, $\mathscr{B} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{W})$. The composition of $\mathscr{A}$ and $\mathscr{B}$ is denoted $\mathscr{A} \circ \mathscr{B}$ and is the transformation from $\mathcal{U}$ to $\mathcal{V}$ defined as

$$(\mathscr{A} \circ \mathscr{B})(\boldsymbol{\alpha}) = \mathscr{A}[\mathscr{B}(\boldsymbol{\alpha})]$$

for all $\boldsymbol{\alpha} \in \mathcal{U}$, -see Figrue 1.5 on page 33.

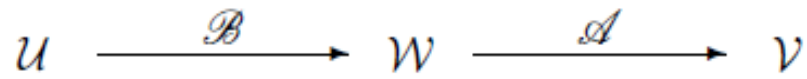



Figure 1.5: The composite transformation $\mathscr{A} \circ \mathscr{B}$

# 1.5 Linear transformation



- Composite transformation
  - The composite transformation is also a linear transformation

Since both $\mathscr{B}$ and $\mathscr{A}$ are linear, we have for all $\alpha_1, \alpha_2 \in \mathcal{U}$, $k \in \mathcal{F}$,

$$\begin{aligned}
(\mathscr{A} \circ \mathscr{B})(\alpha_1 + \alpha_2) &= \mathscr{A}[\mathscr{B}(\alpha_1 + \alpha_2)] = \mathscr{A}[\mathscr{B}(\alpha_1) + \mathscr{B}(\alpha_2)] \\
&= \mathscr{A}[\mathscr{B}(\alpha_1)] + \mathscr{A}[\mathscr{B}(\alpha_2)] = (\mathscr{A} \circ \mathscr{B})(\alpha_1) + (\mathscr{A} \circ \mathscr{B})(\alpha_2) \\
(\mathscr{A} \circ \mathscr{B})(k\alpha) &= \mathscr{A}[\mathscr{B}(k\alpha)] = \mathscr{A}[k\mathscr{B}(\alpha)] \\
&= k\mathscr{A}[\mathscr{B}(\alpha)] = k(\mathscr{A} \circ \mathscr{B})(\alpha)
\end{aligned}$$

Thus, $\mathscr{A} \circ \mathscr{B}$ is additive and homogeneous, and is in $\mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$.

# 1.5 Linear transformation

Z. Song

- Composite transformation

- Composite transformation -linear transformation, what is the matrix representation?

Theorem 1.17 Let $\mathcal{U}$ be $n$-dimensional linear space with the basis $[\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]$, $\mathcal{W}$ be $p$-dimensional linear space with the basis $[\boldsymbol{\gamma}_1 \quad \boldsymbol{\gamma}_2 \quad \dots \quad \boldsymbol{\gamma}_p]$ and $\mathcal{V}$ be $m$-dimensional linear space with the basis $[\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m]$, $\mathscr{B} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{W})$ and $\mathscr{A} \in \mathcal{L}(\mathcal{W} \mapsto \mathcal{V})$, $B \in \mathcal{F}^{p \times n}$ and $A \in \mathcal{F}^{p \times m}$ be the matrix representations of $\mathscr{B}$ and $\mathscr{A}$ respectively, with respect to the corresponding bases. Then the matrix representation of the composite transformation $\mathscr{A} \circ \mathscr{B} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$ is the $m \times n$ matrix $AB$.